PART 1

1. [**3 points**] Spark API provides a method `mapPartitionsToPair`, in Java, or `mapPartitions`, in Python, to transform an RDD (it was used in the WordCount example, and in the homeworks). Briefly describe what the method does. You do not need to write code.

2. [**3 points**] Consider the adaptation of algorithm k-means++ to solve the weighted variant of k-means clustering for a pointset P, where each $x \in P$ has weight $w(x) \geq 0$. Let S be the current set of selected centers at some point during the algorithm. What is the probability that a given point $y \in P - S$ is selected as a next center?

3. [**3 points**] Consider the ϵ-Approximate Frequent Items (ϵ-AFI) problem for a stream $\Sigma = x_1, x_2, \ldots x_n$, with frequency threshold φ, and accuracy parameter $\epsilon \in (0, \varphi)$. For a desired confidence δ, the Sticky Sampling algorithm (n known) uses sampling rate $r = \ln(1/(\delta\phi))/\epsilon$. State and prove a bound on the expected working memory required by the algorithm.

4. [**4 points**] Referring to similarity search

   (a) Define the Range Reporting (RR) problem, and say how it differs from the Near Neighbor Search (NNS) problem.

   (b) Suppose that a kd-tree is used to store a set P of points in $\mathbb{R}^2$. State the bounds on its space requirements and its query time for the RR problem.

PART 2

1. **[6 points]** Consider a set $P \subset \Re$ of N points in $[0, 1]$ and let $k > 1$ be an integer. Assuming that for each $1 \le i \le k$, the segment $[(i-1)/k, i/k] \subset [0, 1]$ contains at least one point, show that $\Phi^{\rm opt}_{\rm kcenter}(P, k) \le 1/k$.

2. **[7 points]** Let $\Sigma = x_1, x_2, \ldots, x_n$ be a stream representing a sequence of transactions of an e-commerce site. Each x_i is a pair (p_i, γ_i), where p_i is a product (from a universe U) and γ_i is a binary flag which indicates if in this transaction p_i has been *purchased* ($\gamma_i = 1$) or *returned* ($\gamma_i = 0$). For a product $p \in U$, define its *net sales* as

$$n_p = |\{(p_i, \gamma_i) : \ 1 \le i \le n, \ p_i = p, \ \gamma_i = 1\}| - |\{(p_i, \gamma_i) : \ 1 \le i \le n, \ p_i = p, \ \gamma_i = 0\}|$$

   (a) Describe how to obtain a $d \times w$ array C of counters which can be employed to provide, for any product p, an unbiased estimate $\tilde{n}_p$ of its net sales n_p. Also, say how $\tilde{n}_p$ is computed.

   (b) How accurate is the estimate $\tilde{n}_p$ when $p_1 = p_2 = \cdots = p_n = p$?

Total time: 2 hours